Quantam.
Based secure
Data transmission
system

Introduction

A Quantum Based Secure Data Transmission System is a method of sending information securely using the principles of quantum physics. The most common approach is called Quantum Key Distribution (QKD), where encryption keys are shared using quantum particles (usually photons).

Quantum communication uses properties such as superposition and quantum particles (like photons) to transmit information securely. Because of these unique q uantum properties, it becomes practically impossible for attackers to copy or measure the data without being noticed.

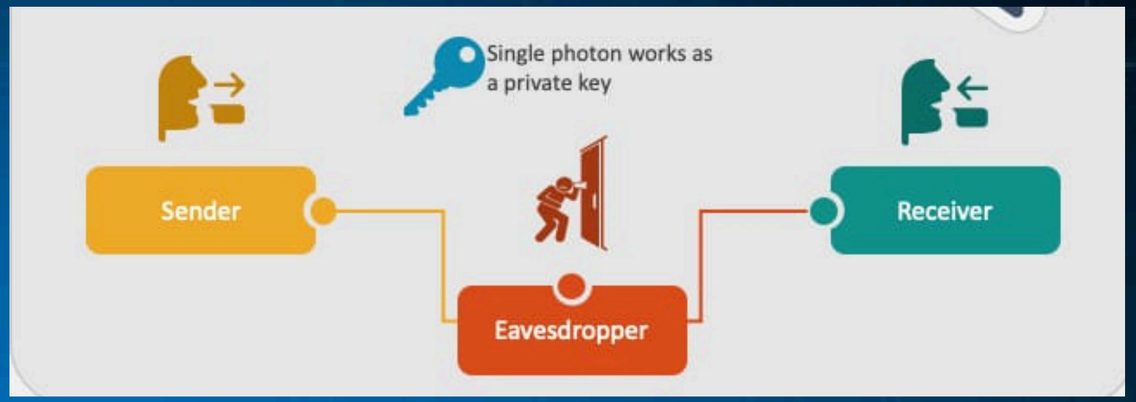
Abstract

This paper x-rays quantum-secure data transmission protocols. Quantum cryptography relies on two fundamental principles of quantum theory; measuring a quantum state disturbs it unless it is an eigenstate of the measurement operator; and it is impossible to create an exact copy of an unknown quantum state. These principles enable secure communication by encoding information using quantum states. The BB84 protocol, the first quantum key distribution (QKD) protocol, was introduced in 1984 by Bennett and Brassard, with Wiesner presenting related ideas about a decade earlier. Another historical protocol is the Einstein-Podolsky-Rosen protocol proposed by Artur Erkert in 1991, which is thought of as an independent invention of Quantum Cryptography using a different route. Over the last two decades, there has been active research in QKD technology, resulting in significant enhancements in performance metrics like transmission distance and key rate for fiber-based QKD systems,

The proposed system aims to demonstrate how quantum principles can be applied to create a secure communication channel that is resistant to hacking and interception. By integrating quantum cryptography techniques with classical communication networks, the system provides a highly reliable method for secure data transfer.

This technology has significant potential applications in fields such as banking, military communication, government networks, and cloud security, where protecting confidential information is essential. Quantum-based secure communication is expected to play a key role in the future of cybersecurity.

Quantum Crytography



Quantum Cryptography is a secure communication method that uses the principles of quantum mechanics to protect information. Unlike traditional cryptography, which relies on complex mathematical algorithms, quantum cryptography uses the physical properties of quantum particles (such as photons) to generate and distribute secure encryption keys.

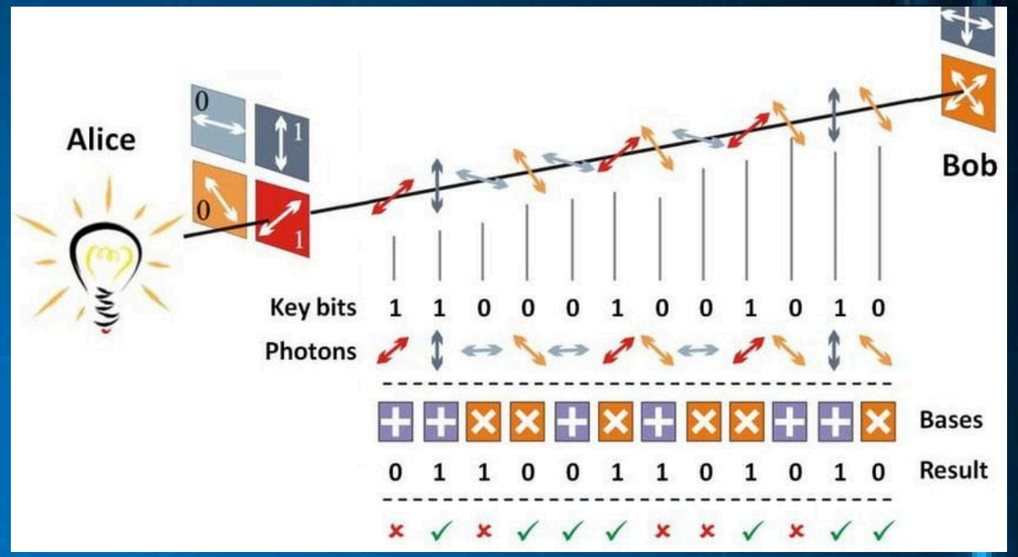
1.Superposition : In quantum mechanics, a particle can exist in multiple states at the same time. This property helps encode information in different quantum states.

2.Heisenberg Uncertainty PrincipleMeasuring a quantum system change its state. If an attacker tries to observe the transmitted quantum particles, the system will detect the disturbance.

3.Quantum Entanglement Two particles can be linked so that the stateof one instantly affects the state of the other, even if they are far apart. This property can be used for secure communication.

BB84 Protocol

The BB84 Protocol is the first and most widely used Quantum Key Distribution (QKD) protocol. It was proposed in 1984 by Charles Bennett and Gilles Brassard.
The main goal of the BB84 protocol is to securely generate a secret cryptographic key between two parties so that they can communicate securely.
The protocol uses quantum mechanics principles, mainly the polarization of photons, to transmit information.



Participants in BB84

1.Alice – The sender who transmits the quantum bits (qubits).
2.Bob – The receiver who measures the qubits.
3.Eve – The eavesdropper who tries to intercept the communication.Polarization Bases Used

System Architecture

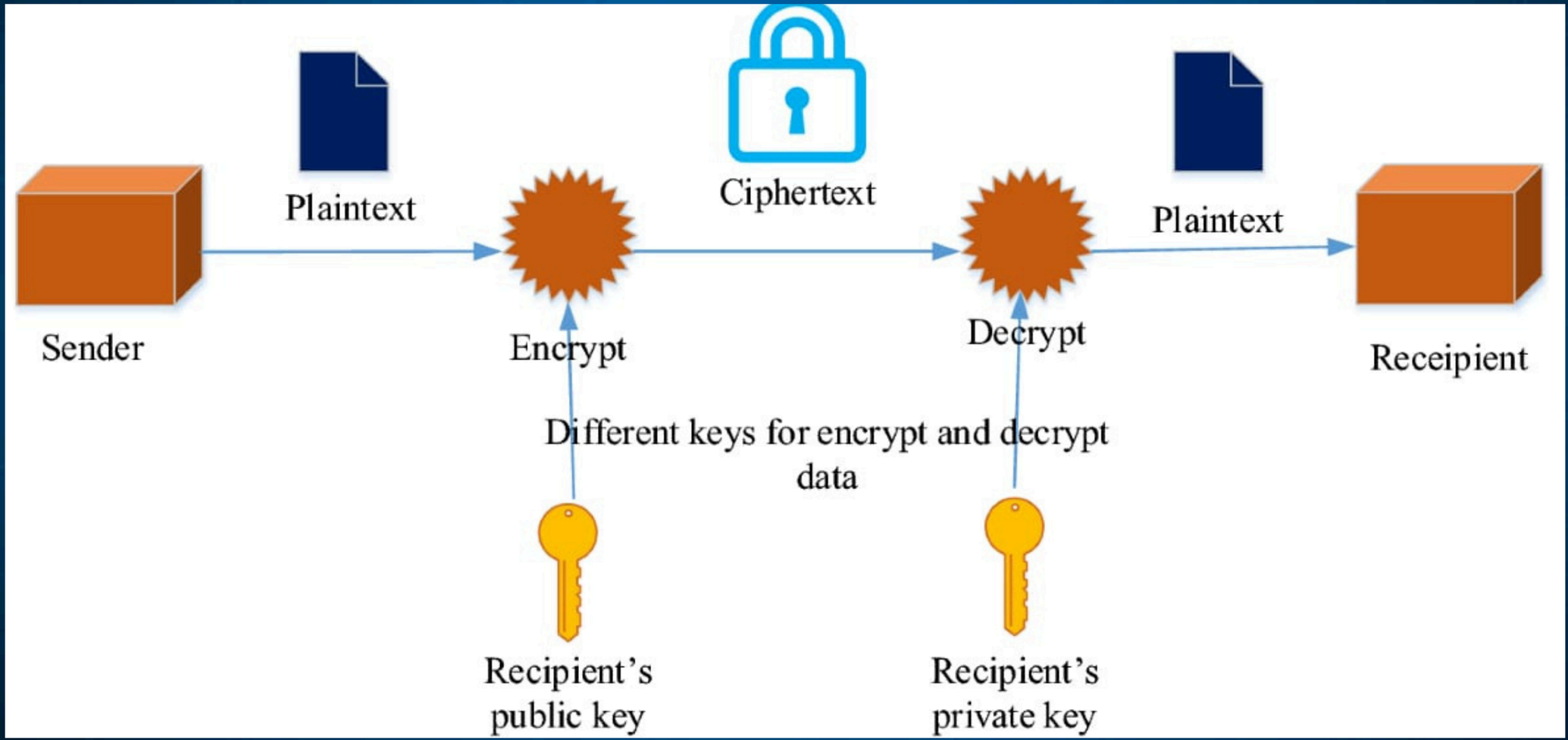
The system architecture of quantum computing describes how different hardware and software components work together to perform quantum computations. Unlike classical computers that use bits (0 or 1), quantum computers use qubits, which can exist in multiple states simultaneously due to superposition.

A quantum computing system is built using several layers that control and process quantum information while maintaining extremely sensitive quantum states.

Main components of System Architecture

1. Quantum Processor
2. Control Electronics
3. Cryogenic System
4. Classical Computer System

Block Diagram



Working Principle of Quantum-Based Secure Data Transmission

- Quantum secure communication uses the principles of quantum mechanics to protect data
- The sender (Alice) first generates a random sequence of bits.
- These bits are encoded into photons (quantum particles) using different polarization states.
- The photons are sent to the receiver (Bob) through a quantum communication channel such as optical fiber or free space.
- Bob measures the received photons using random measurement bases.
- Alice and Bob then compare the bases they used through a public classical channel.
- Only the bits measured with the same bases are kept to form a secret key.
- If an eavesdropper tries to intercept the photons, the quantum state changes, which reveals the attack.
- After checking for errors, Alice and Bob generate a shared secret key.
- This secret key is then used to encrypt and securely transmit data.

Algorithm for Quantum-Based Secure Data Transmission

Step 1: Alice generates a random sequence of bits (0s and 1s).
Step 2: Alice randomly chooses polarization bases (Rectilinear + or Diagonal ×) for each bit.
Step 3: Alice encodes the bits into photons (qubits) according to the selected polarization bases.
Step 4: Alice sends the encoded photons to Bob through a quantum communication channel.
Step 5: Bob randomly selects measurement bases (+ or ×) to measure each received photon.
Step 6: Bob records the measured bits and the bases used.
Step 7:Alice and Bob communicate through a classical public channel and compare the bases they used.
Step 8: They discard the bits where their bases do not match.
Step 9: The remaining bits form the raw secret key.
Step 10: Alice and Bob compare a small portion of the key to check for errors or eavesdropping.
Step 11: If the error rate is low, they perform error correction and privacy amplification.
Step 12: The final result is a secure shared secret key, which is used for encrypted data transmission.

Applications

1.Banking and Financial Security

2.Military and Defense Communication

3. Government and Diplomatic Communication

4.Secure Cloud Storage

5.Satellite Communication

6. Healthcare Data Security

7.Internet and Network Security

Conclusion

Quantum-based secure data transmission provides a highly secure method of communication by using the principles of quantum mechanics. Unlike traditional encryption techniques, quantum cryptography ensures that any attempt to intercept the data can be immediately detected because measuring a quantum state disturbs it.

As digital communication continues to grow, the need for stronger security systems is increasing. Quantum secure communication is expected to play an important role in the future of cybersecurity, especially in areas such as banking, defense, government networks, and cloud computing.

In conclusion, quantum-based secure data transmission offers a promising solution for protecting sensitive information and ensuring safe communication in the modern digital world.

THANK YOU!